1. Geometry

1.1. Similarity

- If two objects are similar, that means that they are the exact same shape but different in size
- All corresponding lengths must be in the same ratio
- **AAA (Angle – Angle – Angle) Rule:** All the corresponding angles of the triangles must be equal.

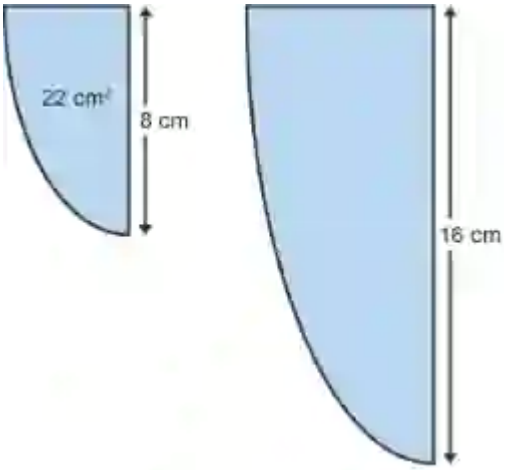
Area and Volume of Similar Shapes

- Similar shapes are enlargements of each other using a scale factor

Measure	Multiplier or Divider
Length	$Scale\ factor$
Area	$(Scale\ factor)^2$
Volume	$(Scale\ factor)^3$

Worked Examples

- Example 1:

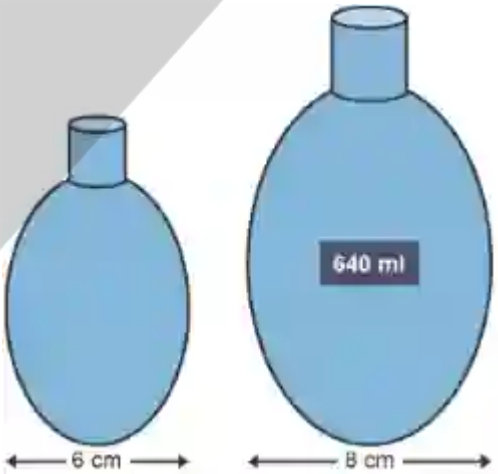


$\text{Small to Big} = \frac{\text{Big}}{\text{Small}} = \frac{16}{8} = 2$

$\text{Scale Factor} = 2$

$\text{Area of Bigger Shape} = 2^2 \times 22 = 88$

Example 2:



$\text{Big to Small} = \frac{\text{Small}}{\text{Big}} = \frac{6}{8} = \frac{3}{4}$

$\text{Scale Factor} = \frac{3}{4}$

$\text{Volume of Smaller Shape} = \left(\frac{3}{4}\right)^3 \times 640 = 270 \text{ ml (to the nearest ml)}$

- Direct Formula to Use

$$\left(\frac{A_1}{A_2}\right) = \left(\frac{L_1}{L_2}\right)^2 \quad \text{and} \quad \left(\frac{V_1}{V_2}\right) = \left(\frac{L_1}{L_2}\right)^3$$

1.2. Geometrical Terms

Congruent

- Two shapes that have the identical shape and size

Scale Factor

- A number that indicates how much a shape has been enlarged or reduced in size

Vertex

- An angular point where two or more lines or edges meet

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Perpendicular Bisector

- A line that bisects a line segment in two equal parts and makes a angle of 90 degrees at the point of intersection

Parallel Lines

- Two or more lines that are always the same distance apart and never intersect

Similar

- They have the **same shape** but **different sizes**.
- All corresponding **angles are equal**.
- All corresponding **side lengths are in the same ratio** (scale factor)

Acute Angle

- An angle **less than 90°**

Obtuse Angle

- An angle **greater than 90° but less than 180°**

Reflex Angle

- An angle **greater than 180° but less than 360°**

Interior Angles

- The angles **inside** a polygon or shape

Exterior Angles

- The angles **outside** a polygon, formed by extending one side.
- For any polygon, **exterior angles add up to 360°**

Right Angle

- An angle of **90°**.
- Shown with a **small square** in diagrams

Bearings

- A **direction** measured:
  - **Clockwise**
  - **From North**
  - Written as a **three-digit number** (e.g., 045°, 120°, 270°)

Perpendicular

- Two lines are **perpendicular** if they meet at a **right angle (90°)**

Parallel

- Two lines are **parallel** if:
  - They are always the **same distance apart**.
  - They **never meet**, no matter how far extended

Line

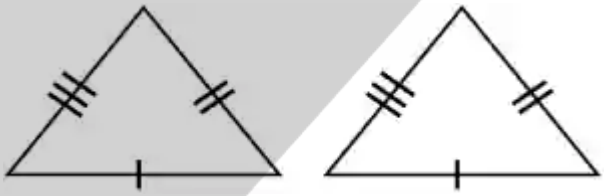
- A **straight path** that:
  - Extends **infinitely** in both directions.
  - Has **no thickness**

Point

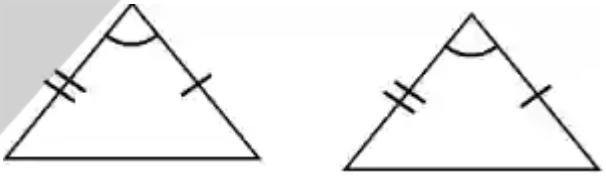
- A precise **location** in space.
- Has **no size**, no length, and no width

1.3. Congruence

- If two objects are congruent, that means that they are identical. Same shape and same size
- **SSS (Side – Side – Side) Rule:** All three sides of the triangles must be equal



- **RHS (Right angle – Hypotenuse – Side) rule :**
- There must be two right-angled triangles
- **SAS (Side – Angle – Side) rule:**



- **ASA (Angle – Side – Angle) rule:** The sides adjacent to the equal angle angles must be of the same length.

Angle Notation:

- Angles are represented with this symbol - $\angle$
- Its written like $\angle ABC$ with the middle letter (B) being the vertex

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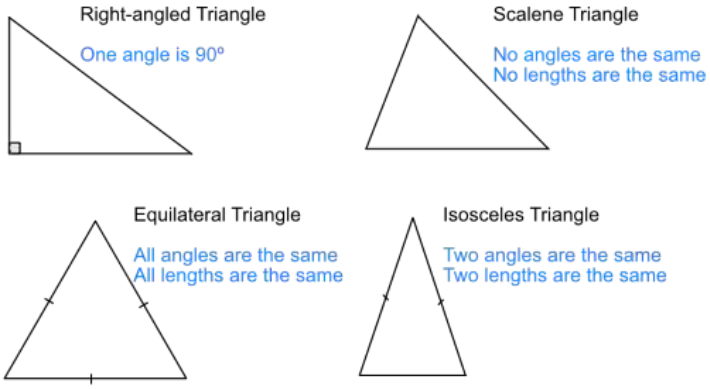
1.4. Triangles



Equilateral Triangle

- All **three sides are equal** in length.
- All **three angles are equal** (each **60°**).
- Perfectly symmetrical.

Isosceles Triangle

- Has **two equal sides**.
- The angles **opposite the equal sides** are also **equal**.
- Has **one line of symmetry**.
- Useful for angle-finding questions using the rule
- Base angles in an isosceles triangle are equal

Scalene Triangle

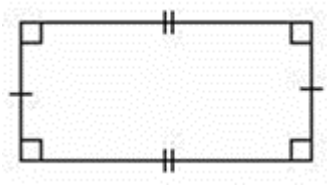
- **No sides are equal**.
- **No angles are equal**.
- Irregular shape with **no symmetry**.

Right-Angled Triangle

- Contains **one right angle (90°)**.
- The side opposite the right angle is the **hypotenuse** (the longest side).
- Used for:
  - **Pythagoras' theorem**
  - **Trigonometry (sin, cos, tan)**

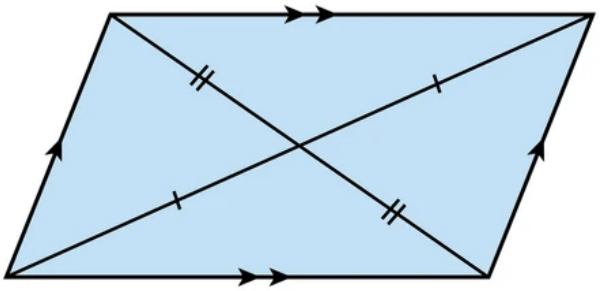
1.5. Quadrilaterals

Rectangle



- Opposite sides are parallel and equal
- All internal angles are 90°
- Diagonals bisect each other
- Diagonals are equal in length

Parallelogram



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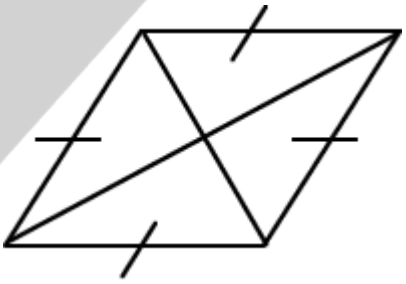
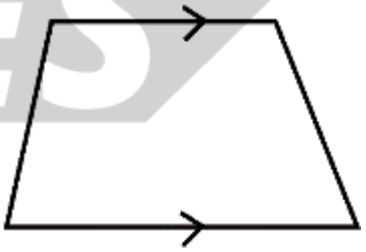
- Opposite sides are parallel and equal
- Opposite angles are equal
- Diagonals are not necessarily equal
- No right angles unless it's a rectangle
- diagonals bisect each other

Rhombus



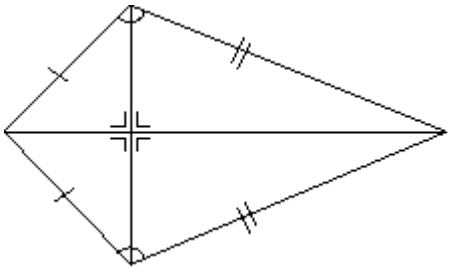
- A parallelogram with all sides equal
- Opposite angles are equal
- Diagonals bisect each other at 90°, and they bisect opposite angles

Trapezium



- One pair of sides is parallel

Kite



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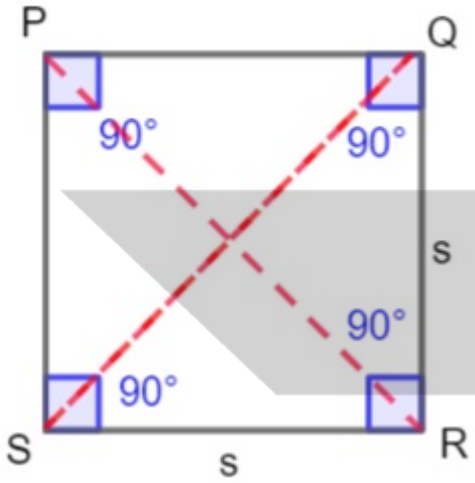
- Two pairs of adjacent sides are equal
- Diagonals are perpendicular to each other
- One pair of opposite angles is equal (those between the unequal sides)
- One diagonal is bisected (the one joining equal angles

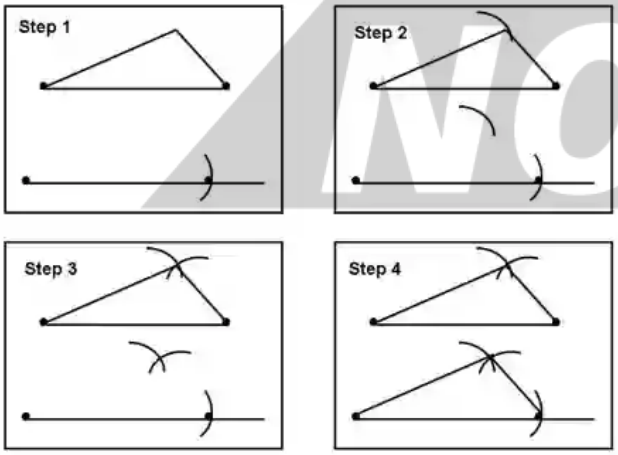
Square



- Four equal sides and four right angles
- Opposite sides are parallel
- Diagonals that are equal in length and bisect each other at right angles
- Diagonals bisect opposite angles

1.6. Construction

- Constructing triangles:



Step 1 : Draw the base line using a ruler

Step 2 : Using a compass, measure the desired length on a ruler and draw an arc

Step 3 : Similarly use the compass to measure the 2nd side and draw an arc

Step 4 : Connect the ends of the baseline to the point where the arc intersect.

Scale Drawings :

1 cm = 100 m	- 1 cm on the grid represents 100 m
1 : 10000	- 1 cm on the graph means 10000 cm in real life
100 m	- the distance marked represents 100 m

- Always make sure to convert to correct units when drawing diagrams

1.7. Symmetry and Nets

- Line of Symmetry: Divides a two-dimensional shape into two congruent (identical) shapes
- Plane of Symmetry: Divides a three-dimensional shape into two congruent solid shapes
- Rotational Symmetry: The number of times a shape looks the same during a 360° rotation about its center

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Shape	Number of Lines of Symmetry	Rotational Symmetry Order
Square	4	4
Rectangle	2	2
Parallelogram	0 (General parallelogram have 0 line of symmetry)	2
Rhombus	2	2
Trapezium	0	1
Kite	1	1
Equilateral triangle	3	3
Regular hexagon	6	6

Nets

- This is a 2D shape that can be folded to form a 3D shape

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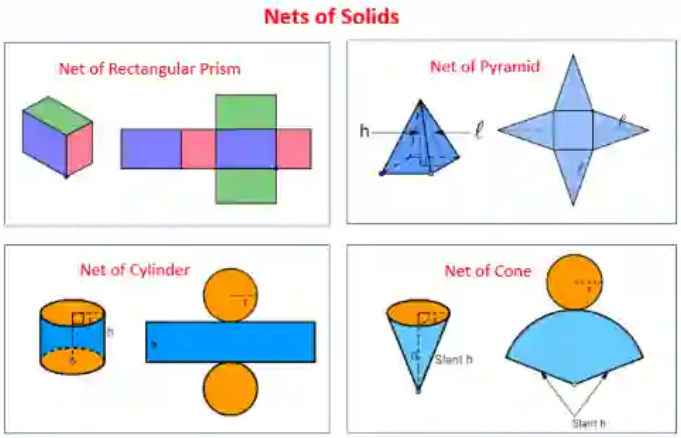
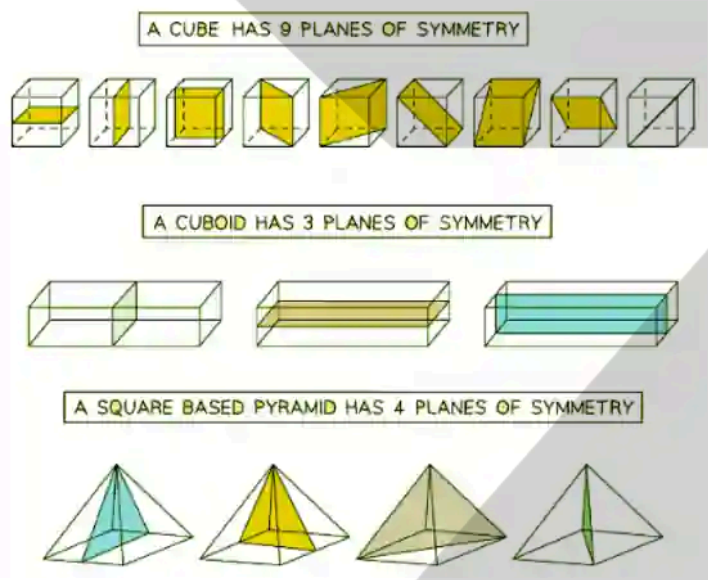


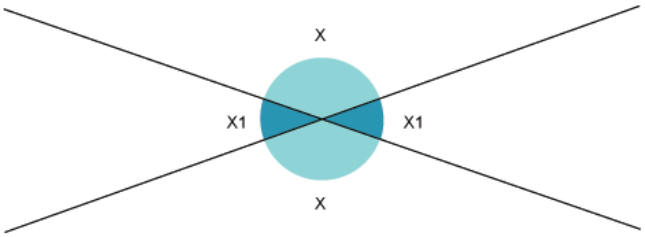
- This is usually used to help find areas of 3D shapes

Symmetry of 3D shapes

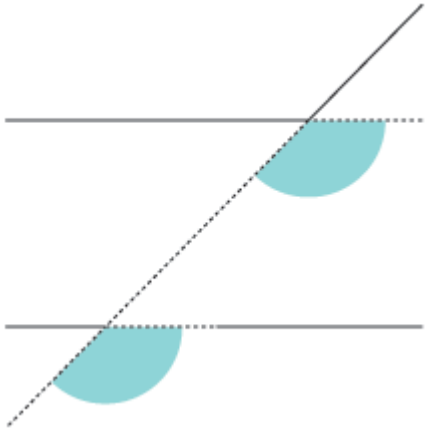


1.8. Angles

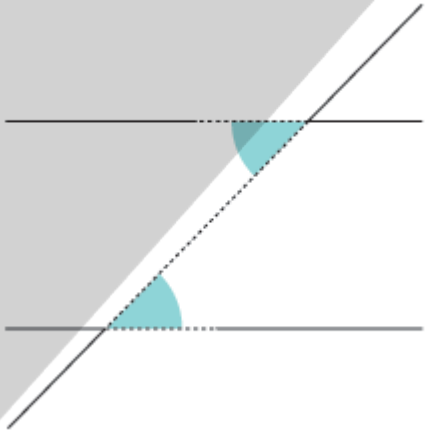
- Sum of angles at a point $= 360°$
- Angles on a straight line $= 180°$
- Sum of angles in a triangle $= 180°$
- For regular polygon
  - Each Exterior Angle of a regular polygon $= \frac{360°}{n}$
  - Each Interior Angle of a regular polygon $= 180° - \frac{360°}{n}$ or $\frac{(n-2)\times 180}{n}$
- Vertically opposite angles are equal



- Corresponding angles are equal

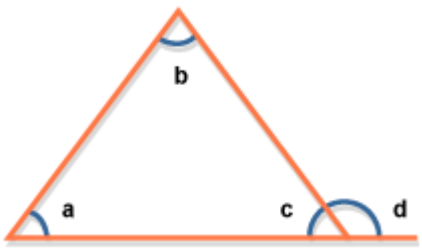


- Alternate angles are equal



- Co-interior angles add up to 180°

- Exterior angle=sum of interior opposite $\angle a + \angle b$



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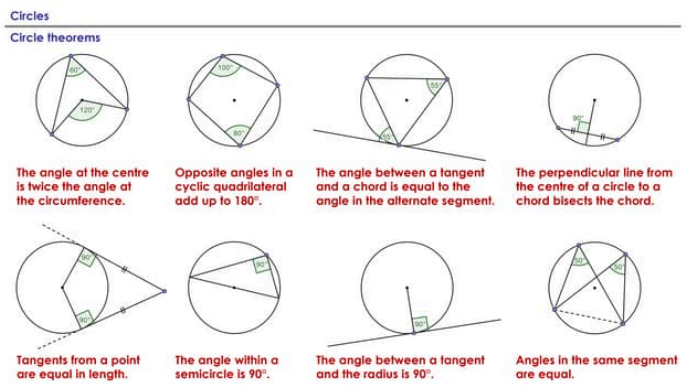
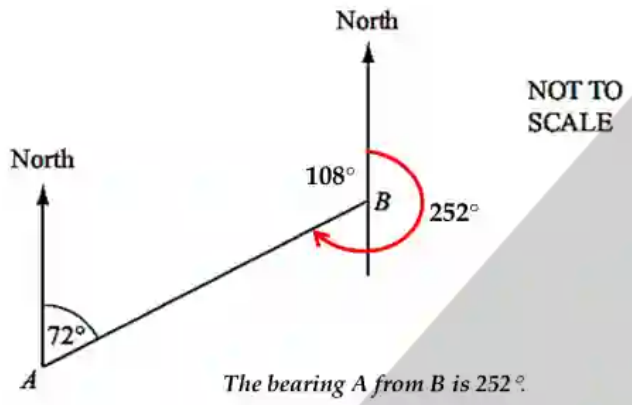
1.9. Circle Theorem



1.10. Bearings

- The bearing of a point B from another point A is:
  - An angle measured from the north at A.
  - In a clockwise direction.
  - Written as three-figure number (i.e. from 000° to 360°)



How to Calculate Bearings (from the example above)

- **Step 1 — Identify the starting point**
  - Bearing of A from B" means:
    - Start at B
    - Look towards A
- **Step 2 — Draw or identify the North line**
  - At point B, draw a vertical line pointing up — this is North.
  - Bearings are always measured from the North line.
- **Step 3 — Bearings must be measured *clockwise***
  - This is the most important rule.
  - Always rotate clockwise from the North line until you reach the line pointing to the other point.
- **Step 4 — Look at the angle given in the diagram**
  - The diagram shows 108°, but this angle is measured anticlockwise from the North line down to line BA.

  BUT bearings are not measured anticlockwise — only clockwise.
- **Step 5 — Use the full circle to find the clockwise angle**

  The full turn around a point is 360°.

  Since the diagram gives the anticlockwise angle:

  $$\text{Clockwise angle} = 360^\circ - 108^\circ = 252^\circ$$
- Final Answer = 252

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